

ALPINE ZONE

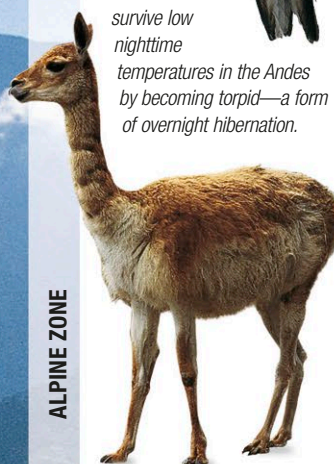
FOREST ZONE

LOW ELEVATION



MOUNTAIN HUMMINGBIRDS

survive low nighttime temperatures in the Andes by becoming torpid—a form of overnight hibernation.



THE VICUNA'S blood is especially good at gathering oxygen at high elevations.



THE NORTHERN PUDU *is the smallest species of deer and occurs across Andean mountain habitats—from montane rainforests up to paramo grassland.*



GELADAS are ground-living monkeys allied to baboons. They are confined to high elevations of Ethiopia, where they graze on grassland plateaus.



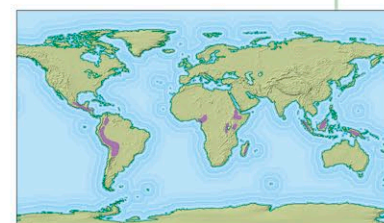
THE EASTERN GORILLA *uses the dense vegetation of the forest zone for both food and cover.*

Tropical mountains

In the tropics, the generally warm climate means that mountain vegetation zones extend much higher than they do in montane habitats elsewhere in the world. For example, near the equator, trees often grow at elevations of up to 13,200 ft (4,000 m), which is why many tropical mountains are forested to their summits. Above this elevation is the tropical alpine zone, an open landscape dominated by grass and some highly specialized plants. This zone is often above the clouds, which means that nights are cold and frosty and yet the sunshine is fierce.

Many tropical animals have successfully adapted to life at high elevation. They include the vicuna, which can be found up to 18,100ft (5,500m) in the South American Andes, and the yak, which reaches a record 19,800ft (6,000m) just north of the tropics in the Himalayas. Birds also live at great heights: in South America, for example, mountain hummingbirds called Andean hillstars often feed at over 13,200ft (4,000m). The Andean hillstar's minute size means it has difficulty storing enough energy to enable it to survive the cold nights. To combat the problem, its nocturnal heartbeat slows down and its temperature plummets, conserving energy.

The cloud-covered forest below the alpine zone is the habitat of some of the world's most endangered animals. They include the eastern gorilla—a species restricted to the mountains of central Africa—and the resplendent quetzal, a bird that lives in the cloud forests of Central America. The abundant moisture means that the forest zone also teems with many different species of frogs, living both on the ground and in trees.



TROPICAL MOUNTAINS include a major chain, the Andes, and isolated ranges in East Africa and Southeast Asia.

MINING AND QUARRYING

Once isolated from human activities, mountain wildlife now faces many changes. The greatest of all is global warming, which alters the natural pattern of vegetation zones. Some mountain animals can benefit from this, but others face steadily shrinking habitats, because they live in a narrow range of elevations, or on isolated ranges or peaks. More

specific threats to mountain animals include hydroelectric projects, and the rapid expansion of mines and quarries. Dust, pollution, and the dumping of waste can all have a harmful effect on mountain wildlife.



ROCK REMOVAL

In some parts of the world, quarrying has destroyed important wildlife habitats. As road building and construction projects continue to increase, the demand for rock continues to grow.